

VAMSI KRISHNA



[Email Address]



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My Portfolio

Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science and Engineering (Information Security)

Sept 2024 – July 2027

Vellore, Tamil Nadu

Experience

Software Developer Intern

AIAT India

July 2025 – Nov 2025

Hyderabad

- Monitored and maintained cloud-hosted environments using Linux and AWS services(S3 and IAM), ensuring smooth deployment workflows.
- Utilized Git for effective source code management while contributing to internal tool development in a team-based workflow.
- Designed and implemented web-based systems using React and Node, building responsive interfaces with Tailwind.

IEEE-VIT Student Chapter

Project Head

May 2024 – Present

Vellore, Tamil Nadu

- Engineered and launched a national-level conference platform using React and Tailwind, improving performance by 30 percent and reaching 1,000+ participants.
- Developed Python CLI tooling with OOP design (Click library) for automated speaker onboarding and submission validation, enforcing metadata standards and reducing setup friction by 50% for 200+ Participants.
- Implemented data governance on AWS S3-backed storage: lineage tracking, access boundaries, and upload guardrails, preventing cross-path issues in multi-team workflows.
- Built Databricks-inspired monitoring dashboards for platform usage (job runtime, compute costs via boto3), delivering visibility to leadership and cutting overspend by 25%.

Projects

Quantum-safe Image Signing System | React Native, Python, FastAPI, Dilithium2

Sept 2025 – Oct 2025

- Built a quantum-safe image signing system using React Native frontend and FastAPI backend with the post-quantum Dilithium2 signature algorithm and liboqs Quantum safe Library for the core functionality.
- Implemented secure end-to-end workflows for signing and verification of images.

Agentic Knowledge-Base Service | Python ,Langchain, Chroma DB

Aug 2025 – Oct 2025

- Developed a knowledge-base agent powered by Retrieval-Augmented Generation (RAG) and long-context AI Embedding models.
- Implemented persistent vector storage and document chunking using Chroma DB.
- Enhanced retrieval accuracy by 40 percent in long-context reasoning tasks via late-chunking optimization.

Skills

Languages: Python, C++, Go, Shell, SQL, JavaScript, Java

Frameworks: Numpy, Pytorch, Node.js, PennyLane

Softwares & Tools: Git, GitHub, AWS, Linux

Achievements

Quantum Computing: Built quantum simulation workflows with Qiskit and Mitiq (contributed pull requests to Mitiq), implementing resource estimation and error mitigation for benchmarking noisy circuits with NumPy/SciPy linear algebra.

Hackathons: Secured **2 First-Place wins** and participated in **10+ national-level hackathons**, consistently contributing to full-stack and AI-driven solutions.

Amazon ML School Hack 2025: Ranked among the **Top 40 teams in India** for developing an end-to-end Dynamic ML –based solution under competitive timed constraints.